Graph-Based Entity Resolution and Completion for Academic Knowledge Graphs

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Academic Knowledge Graphs

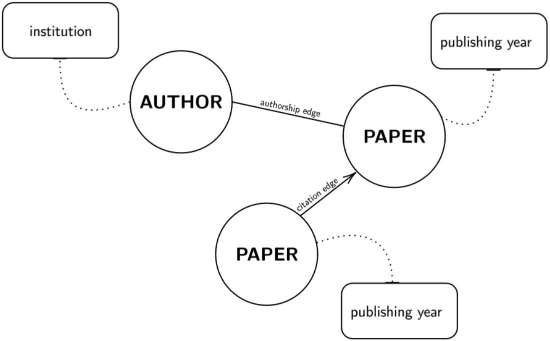
Definition

Knowledge graphs encode and integrate varied types of data into a unified framework. The value of academic knowledge graphs lies in their ability to link diverse entities and reveal complex relationships, which supports advanced bibliometric analyses.

Examples of the aforementioned concepts are as follows:

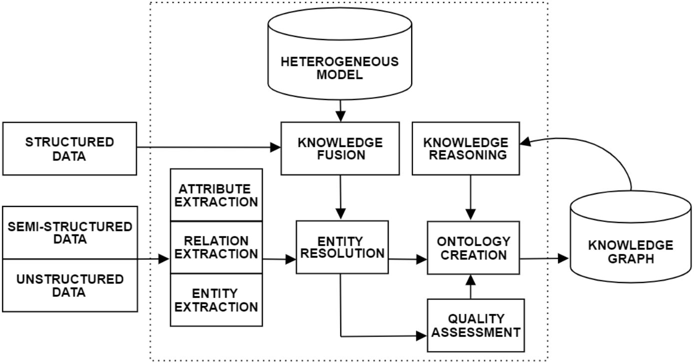
- authors & papers (*entities*)
- authorship & citation (*relationships*)
- affiliations & publishing year (*bibliometric data*)
- author disambiguation & citation analysis (*advanced analyses*)

Topology


institution
AUTHOR
authorship edge
PAPER
publishing year
citation edge
PAPER
publishing year

Methodology

DATA COLLECTION → DATA INTEGRATION → ENTITY RESOLUTION → GRAPH CONSTRUCTION → GRAPH ENRICHMENT



Entity Resolution

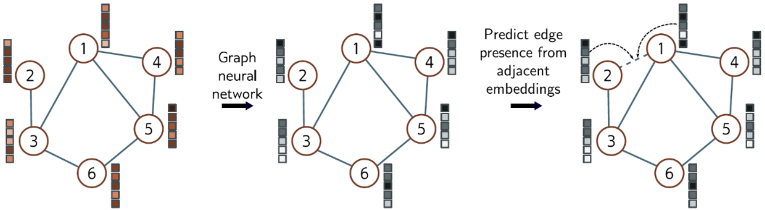
Definition

Entity resolution (also known as record linkage or deduplication) is the process of identifying and linking records that refer to the same real-world entity across different datasets.

Techniques for entity resolution include:

- Rule-Based Approaches
- Probabilistic Models
- Machine Learning Methods

Link Prediction for Graph Completion



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Properties

Rows of Metadata: 720,535

Author-Paper Pairs: 2,571,821

Citing Papers: 693,706

Cited Papers: 625,223

Components

Citation Network

citing_doi	cited_doi
10.1103/PhysRevSeriesI.11.215	10.1103/PhysRevSeriesI.1.1
10.1103/PhysRevSeriesI.12.121	10.1103/PhysRevSeriesI.1.166
10.1103/PhysRevSeriesI.7.93	10.1103/PhysRevSeriesI.1.166
10.1103/PhysRevSeriesI.16.267	10.1103/PhysRevSeriesI.2.35
10.1103/PhysRevSeriesI.17.65	10.1103/PhysRevSeriesI.2.112

Metadata

Feature	Description
ID	Unique identifier for the article
Title	Title of the scholarly article
Publisher	Name of the publisher
Journal	Journal in which the article was published
Issue	Issue number of the journal
Volume	Volume number of the journal
Page Start	Starting page number of the article
Page End	Ending page number of the article
Sequence Number	Sequence number of the article
Date	Date when the article was published
Number of Pages	Total number of pages in the article
Article Type	Type of the article (e.g., research article, review)
Identifiers	Unique identifiers for the article (e.g., DOI)
Rights	Rights associated with the article
Authors	Names of authors contributing to the article
Affiliations	Institutions to which authors are affiliated
Has Article ID	Indicates whether the article has an ID
Classification Schemes	Classification schemes associated with the article

Exploratory Data Analysis

Citation Network

	citing_doi	cited_doi
Unique Count	693,706	625,223
Most Frequent	10.1103/RevModPhys.83.471	10.1103/PhysRevLett.77.3865
Mean	14.17	15.73
Standard Deviation	12.69	53.70
Maximum	607	14,357

Metadata

Journal ID	Journal Name	Entry Count
PRB	Physical Review B	207,609
PRL	Physical Review Letters	133,694
PRD	Physical Review D	103,017
PRA	Physical Review A	88,357
PRE	Physical Review E	65,626
PR	Physical Review	47,940
PRC	Physical Review C	43,842
PRAPPLIED	Physical Review Applied	5,295
PRRESEARCH	Physical Review Research	4,8 13
PRMATERIALS	Physical Review Materials	3,951
PRFLUIDS	Physical Review Fluids	3,524
RMP	Reviews of Modern Physics	3,494
PRSTAB	Special Topics - Accelerators and Beams	2,399
PRX	Physical Review X	2,352
PRAB	Physical Review Accelerators and Beams	1,566
PRI	Physical Review (Series I)	1,469
PRPER	Physical Review Physics Education Research	693
PRXQUANTUM	PRX Quantum	497
PRSTPER	Special Topics - Physics Education Research	368
PRXENERGY	PRX Energy	29

Creation of Data Tables

author_table

Field	Details
Identifier	0
DOI	10.1103/PhysRev.1.124
Type	Person
Name	Lachlan Gilchrist
First Name	Lachlan
Surname	Gilchrist
Year	1913
Affiliation	Centro de Física de Materiales, San Sebastián, Spain Donostia International Physics Center, San Sebastián, Spain Physics Department, Technical University of Munich, Garching, Germany
Countries	['Germany', 'Spain']

paper_table

Identifier	DOI
0	10.1103/PhysRevSeriesI.11.215
1	10.1103/PhysRevSeriesI.12.121
2	10.1103/PhysRevSeriesI.7.93
...	...
709802	10.1103/PhysRevFluids.7.093104

Data Quality

Dataset is curated by the journal editors.

Incomplete records and misclassified information are minimized.

Analyses are still constrained by the dataset itself.

Limitations due to potential inaccuracies and missing data.

Contents

Analyses for Author Grouping Strategies

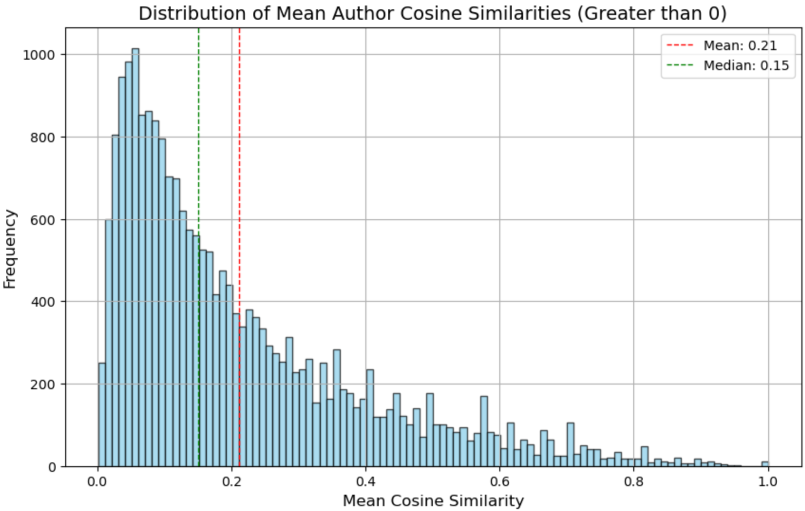
Pairwise Comparisons

1. Authors grouped by ______, and authors with fewer than two papers are discarded.
2. For each author, citation network is referenced to retrieve all citations associated with author's DOIs.
3. Matrix created with authored papers as rows and cited papers as columns.
4. Matrix is filled with binary indicators of the presence of a citation.

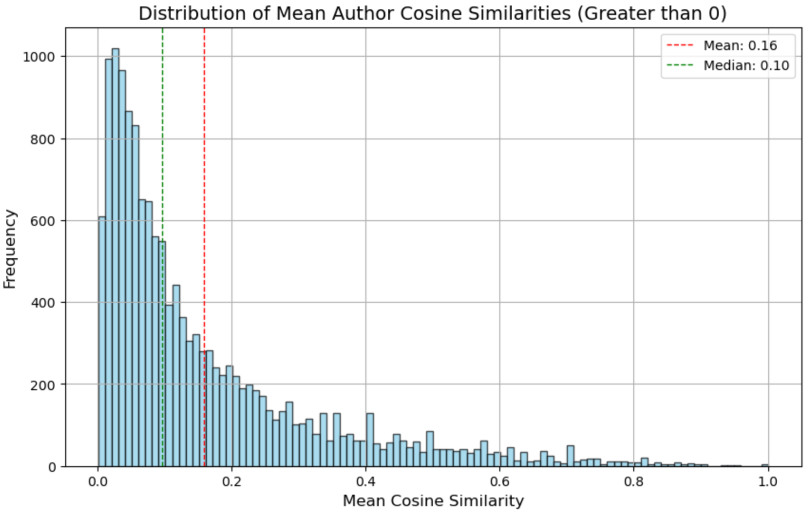
Citation Probabilities

1. Compute cosine similarity matrices to measure the similarity between citation patterns.
2. $\text{cos\_sim}(v_1, v_2) = \frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|}$
3. Examining the distribution of mean cosine similarities, we identify the average citational overlap between authors' papers.
4. Helps in understanding the extent of self-citation or citation of closely-related work.

Full-Name Matching



Surname Matching



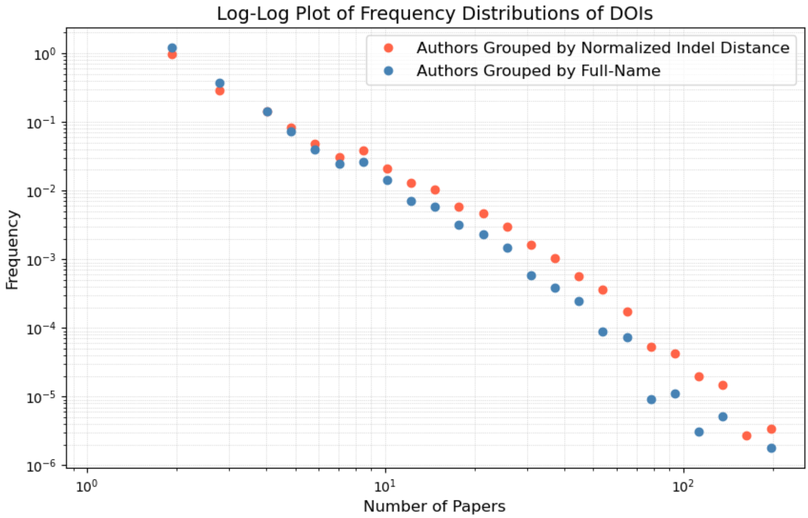
Normalized Indel Distance

Measures the number of **insertions or deletions** required to change one string into another, normalized by the length of the strings.

We set a distance threshold of **80 out of 100** to identify and merge author names with minor discrepancies.

Method is employed with the intention of mitigating issues arising from **typographical errors or naming variations**.

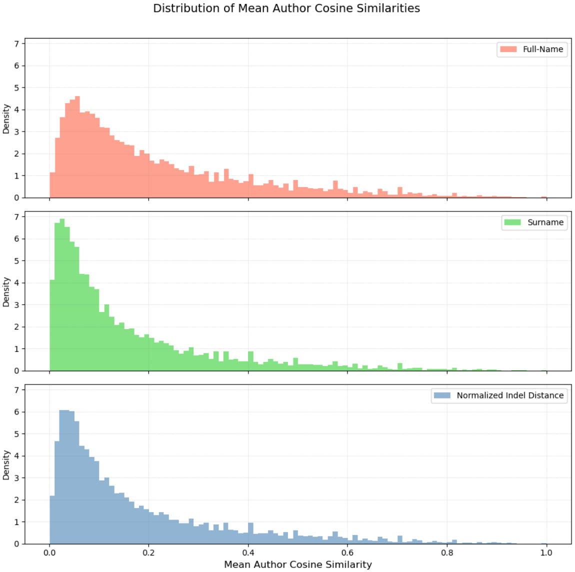
Power Law Relationship



Author Grouping Strategy Statistics

Grouping Method	Author Count	Mean # of Papers	Mean Cos Sim
Full-Name	30,659	4.65	0.21
Surname	20,582	8.21	0.16
Normalized Indel Distance	24,390	6.55	0.18

Comparison of Author Grouping Strategies



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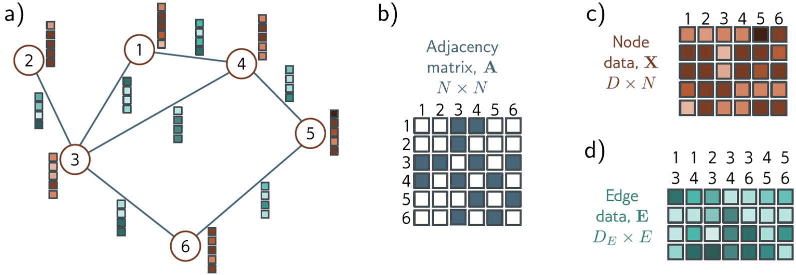
Foundation


a)
1
2
3
4
5
6
b)
Adjacency matrix, $\mathbf{A}$
$N \times N$
1 2 3 4 5 6
1
2
3
4
5
6
c)
Node data, $\mathbf{X}$
$D \times N$
1 2 3 4 5 6
d)
Edge data, $\mathbf{E}$
$D_E \times E$
1 1 2 3 3 4 5
3 4 3 4 6 5 6

Graph Construction

Heterogeneous Data Object Creation

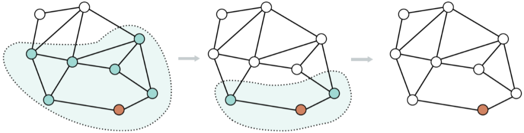
Statistic	Value
Number of 'paper' Nodes	139,966
Number of 'author' Nodes	192,814
Paper Feature Shape	(139966, 8)
Author Feature Shape	(192814, 22)
Citation Edge Index Shape	(2, 700,250)
Authorship Edge Index Shape	(2, 192,392)
Reverse Authorship Edge Index Shape	(2, 192,392)

Edge-Level Splits

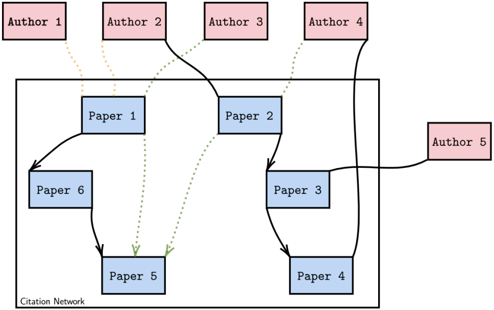
Aspect	Details	Notes
Splitting Method	`RandomLinkSplit` from PyTorch Geometric	Ensures balanced distribution of edges and nodes
Validation Ratio	10%	Predefined ratio for validation edges
Test Ratio	10%	Predefined ratio for test edges
Disjoint Training Ratio	0.30	Reduces overlap between training, validation, and test sets to avoid overfitting
Negative Sampling Ratio	2.0	For every positive edge, 2 negative edges are included
Training Set	Contains positive edges only	Focuses on learning patterns of actual relationships
Validation Set	Includes negative samples	Used for tuning hyperparameters and monitoring performance
Test Set	Includes negative samples	Used for final evaluation of the model

Mini-Batch Loading

Mini-batch loading in GNNs using `LinkNeighborLoader` leverages stochastic gradient descent (SGD) by creating small, manageable mini-batches. It samples 20 neighbors at the first layer and 10 at the second layer, with a batch size of 128.

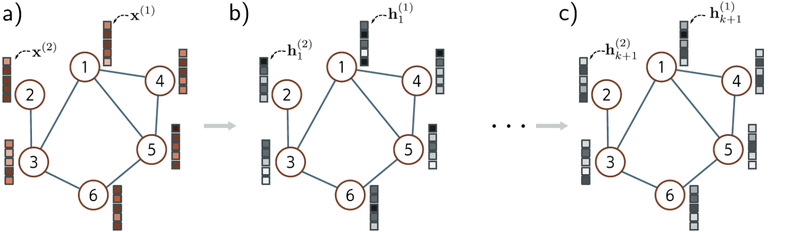


GNN Model Definition



The model comprises four `SAGEConv` layers with a sum aggregation, each followed by a ReLU activation function except the final layer. The architecture is designed to capture multi-hop neighborhood information, enhancing the model's ability to learn complex patterns from the graph structure. The use of ReLU activation functions aids in introducing non-linearity, which is crucial for learning intricate relationships within the data.

Parameter Sharing & Information Aggregation



$$\mathbf{h}_1^{(n)} = \mathbf{a}\left[\boldsymbol{\beta}_0 + \boldsymbol{\Omega}_0 \mathbf{x}_1^{(n)} + \boldsymbol{\Omega}_0 \mathbf{agg}[n]\right]$$

$$\mathbf{h}_{k+1}^{(n)} = \mathbf{a}\left[\boldsymbol{\beta}_k + \boldsymbol{\Omega}_k \mathbf{h}_k^{(n)} + \boldsymbol{\Omega}_k \mathbf{agg}[n]\right]$$

Classifier, Heterogeneous Transformation, Forward Pass

Classifier

Predicts edges between papers and authors.
Uses dot product of node features.
Suitable for link prediction tasks.

Heterogeneous Transformation

Integrates GNN with linear transformations of node features.
Transforms input features to 64-dimensional hidden space.
Transforms model for heterogeneous graphs by adding more weights.

Forward Pass

Transforms paper and author features using linear layers.
Applies GNN to propagate information.
Classifier predicts edges based on transformed features.

Training

Training Steps

Iterate through mini-batches.

Compute predictions and losses.

Perform backpropagation and parameter optimization.

Per-Epoch Activity

Evaluate model performance on validation data.

Adjust hyperparameters based on validation loss.

4 epochs to prevent overfitting.

Trainable Parameters

Learnable weights in linear layers.

Feature-extracting filters in convolutional layers.

77,408 total parameters, all trainable.

Evaluation

Training and Validation Loss per Epoch

Epoch	Training Loss	Validation Loss
001	0.3835	0.2305
002	0.3633	0.2266
003	0.3617	0.2253
004	0.3624	0.2236

Confusion Matrix for Test Data

	Predicted Negative	Predicted Positive
Actual Negative	36,628	1,850
Actual Positive	869	18,370

Classification Report for Test Data

	Precision	Recall	F1-Score
Negative	0.98	0.95	0.96
Positive	0.91	0.95	0.93

Overall Accuracy: 0.95 & Area Under the Curve: 0.98

Contents

Conclusion

Analyses

Constructed an academic knowledge graph from APS data.
Developed author grouping strategies and citational overlap analysis.
Designed a GNN model using `SAGEConv` layers for link prediction.

Results

Author grouping strategy influences the extent of citational overlap.
High accuracy achieved in link prediction for graph completion.
Model uncovers latent relationships between authors and papers.

Improvements

Integrate more detailed paper and author attributes.
Explore alternative GNN architectures.
Develop more sophisticated training techniques.

Thank you very much for your attention.

What questions can I answer?